

1D and 2D List

```
arr = [1, 2, 3, 4, 5]
print(type(arr))

arr1 = [[1, 2], [3, 4], [5, 6]]
print(type(arr1))

print(arr + arr1)

arr[1] = 10
arr[3:] = [40, 50]

arr1[0][1] = 20
arr1[1][1] = 30

print(arr1)

<class 'list'>
<class 'list'>
[1, 2, 3, 4, 5, [1, 2], [3, 4], [5, 6]]
[[1, 20], [3, 30], [5, 6]]
```

List Input From User

```
a = list(map(int, input().split()))
print(a)
print(type(a))

13
[13]
<class 'list'>
```

Array Using Array Module

```
import array as arr

a = arr.array('i', [1, 2, 3, 4, 5])

print(a)
print(len(a))

array('i', [1, 2, 3, 4, 5])
5
```

Sum of Array Elements Using NumPy

```
import numpy as np

a = np.array(list(map(int, input().split())))
print(np.sum(a))

1 2 3 4 5 6 7 8 9
45
```

Sum of Array Elements Without NumPy

```
b = list(map(int, input().split()))

total_sum = 0

for num in b:
    total_sum += num
```

```
print(total_sum)
```

```
1 2 3 4 5 6 7
28
```

✓ Min and Max of Array Using NumPy

```
import numpy as np

a = np.array(list(map(int,input().split())))
print(np.min(a))
print(np.max(a))
```

```
1 3 2 5
1
5
```

✓ Min and Max Without Inbuilt Functions

```
c = list(map(int, input().split()))

mini = maxi = c[0]

for num in c:
    if num > maxi:
        maxi = num

    if num < mini:
        mini = num

print(mini)
print(maxi)
```

```
1 2 3 5 7
1
7
```

✓ Second Largest Using Sorting

```
a = [3,2,4,2,5,6]

print(sorted(a))
print([a[-2]])
```

```
[2, 2, 3, 4, 5, 6]
[5]
```

✓ Second Largest

```
a = list(map(int, input().split()))

for i in range(0,len(a)):
    for j in range(i+1, len(a)):
        if a[i] < a[j]:
            a[i], a[j] = a[j], a[i]

print(a[1])
```

```
13 15 7
13
```

✓ Second Largest Without Sorting

```
a = list(map(int, input().split()))

largest = float('-inf')
second = float('-inf')

for num in a:
    if num > largest:
        second = largest
        largest = num
    elif num > second and num != largest:
        second = num

print(second)
```

```
13 15 7
13
```

Remove Duplicates

```
nums = [2, 2, 1, 4, 5, 1, 6, 6, 6, 4]

seen = {}

for num in nums:
    if num not in seen:
        seen[num] = True

print(sorted(seen))
```

```
[1, 2, 4, 5, 6]
```

Factorial

```
n = int(input())

result = 1

for i in range(n,1,-1):
    result = result * i

print(result)
```

```
13
6227020800
```

Count Frequency of Elements

```
nums = [2,2,1,4,5,1,6,6,6,4]

snums = sorted(nums)

count = {}

for num in snums:
    if num not in count:
        count[num] = 1
    else:
        count[num] += 1

print(count)
```

```
{1: 2, 2: 2, 4: 2, 5: 1, 6: 3}
```

Merge Two Arrays Using NumPy

```
import numpy as np
```

```
list1 = np.array([1,2,4])
list2 = np.array([1,3,4])

list3 = np.concatenate((list1,list2))
list3 = np.sort(list3)

list3 = list3.tolist()

print(list3)

[1, 1, 2, 3, 4, 4]
```

✓ Three Sum

```
class Solution:
    def threeSum(self, nums, target):

        for i in range(len(nums)):
            for j in range(i+1,len(nums)):
                for k in range(j+1,len(nums)):

                    if nums[i] + nums[j] + nums[k] == target:
                        return [i,j,k]
```

✓ Two Sum

```
nums = [2,7,11,15]
target = 9

seen = {}

for i,num in enumerate(nums):

    complement = target - num

    if complement in seen:
        print(seen[complement],i)
        break

    seen[num] = i

0 1
```

✓ Array Rotation

```
arr = [1,2,3,4,5]

k = int(input())

n = len(arr)

arr1 = [0] * n

for i in range(n):
    arr1[(i+k)%n] = arr[i]

print(arr1)

13
[3, 4, 5, 1, 2]
```

✓ Merge Two Sorted Arrays

```
a = [2,5,8,9]
b = [3,4,7,8]
```

```

c = [0] * (len(a)+len(b))

i = j = k = 0

while i < len(a) and j < len(b):

    if a[i] <= b[j]:
        c[k] = a[i]
        i += 1
    else:
        c[k] = b[j]
        j += 1

    k += 1

while i < len(a):
    c[k] = a[i]
    i += 1
    k += 1

while j < len(b):
    c[k] = b[j]
    j += 1
    k += 1

print(c)

```

```
[2, 3, 4, 5, 7, 8, 8, 9]
```

Union of Two Sorted Arrays

```

a = [2,5,8,9]
b = [3,4,7,8]

c = [0] * (len(a)+len(b))

i = j = k = 0

while i < len(a) and j < len(b):

    if a[i] < b[j]:
        c[k] = a[i]
        i += 1

    elif a[i] > b[j]:
        c[k] = b[j]
        j += 1

    else:
        c[k] = a[i]
        i += 1
        j += 1

    k += 1

print(c[:k])

```

```
[2, 3, 4, 5, 7, 8]
```

Intersection of Two Arrays

```

a = [2,5,8,9]
b = [3,4,7,8]

visited = [0] * len(b)

c = [0] * min(len(a),len(b))

k = 0

for i in range(len(a)):
    for j in range(len(b)):

```

```
        if a[i] == b[j] and visited[j] == 0:
            c[k] = a[i]
            k += 1
            visited[j] = 1
            break

result = c[:k]

print(result)
```

[8]